



DATA WAREHOUSE AND ETL PIPELINE IMPLEMENTATION

ID/X Partners - Data Engineer Project Based Internship

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PROJECT OVERVIEW

- Background: The client needs a centralized Single Source of Truth for banking transactions.
- Problem Statement: Data is fragmented across CSV, Excel, and SQL, causing reporting delays.
- Solution: Build a Star Schema DWH and an automated ETL pipeline using Pentaho.



DATA WAREHOUSE CREATION

Strategy: Developing a robust Star Schema to ensure data integrity and query performance.

Key Actions:

- Created the DWH database and defined table structures.
- Implemented Primary Keys and Foreign Keys for relational consistency.
- Applied PascalCase naming conventions for all columns.

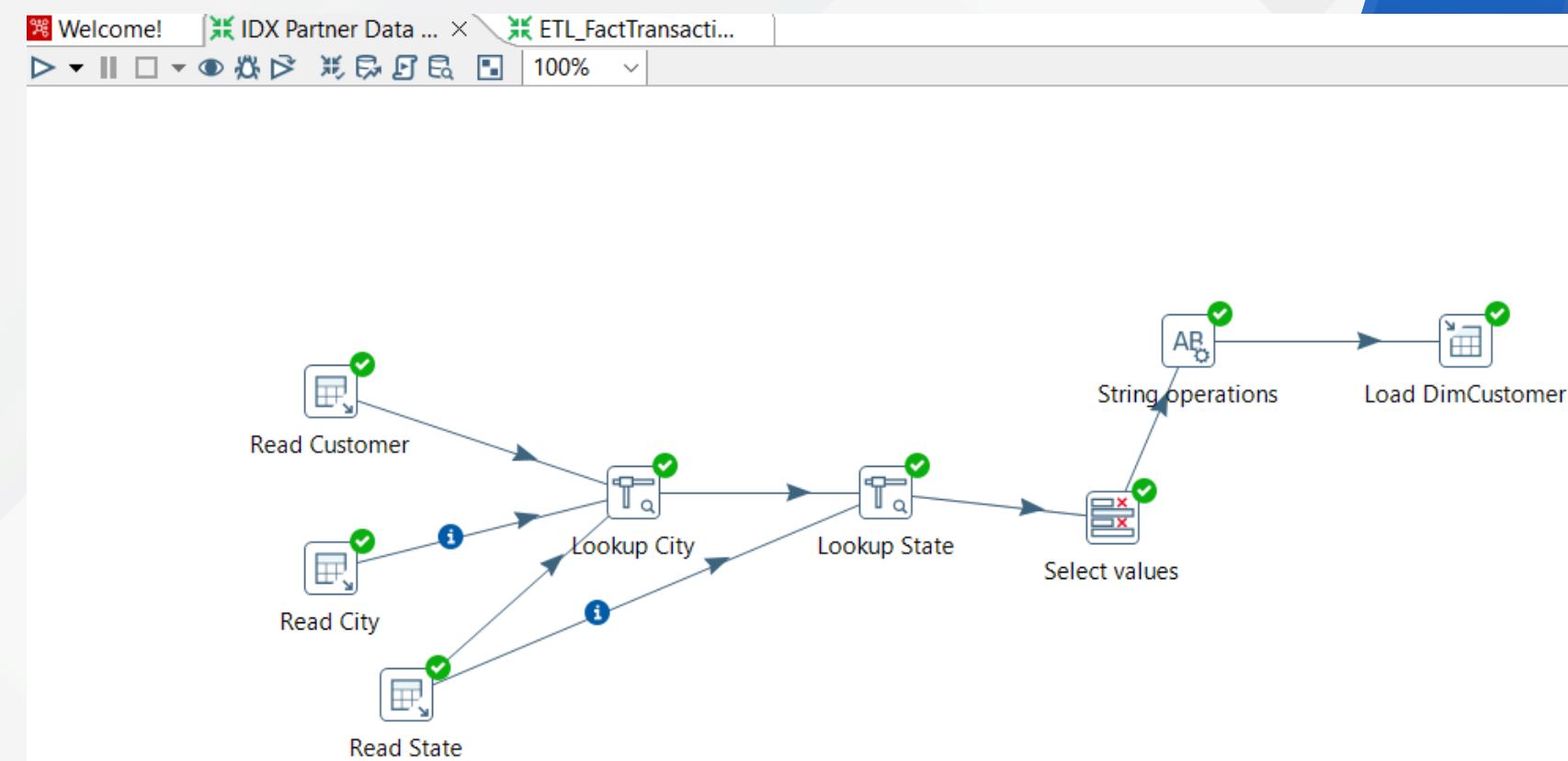
```
SQLQuery1.s...yosee (84))* ✕
1  CREATE DATABASE DWH;
2  GO
3
4  USE DWH;
5  GO
6
7  CREATE TABLE DimCustomer (
8      CustomerID INT PRIMARY KEY,
9      CustomerName VARCHAR(255) NOT NULL,
10     Address VARCHAR(255),
11     CityName VARCHAR(100),
12     StateName VARCHAR(100),
13     Age INT
14 );
15
16 CREATE TABLE DimAccount (
17     AccountID INT PRIMARY KEY,
18     CustomerID INT,
19     AccountType VARCHAR(50),
20     Balance DECIMAL(18, 2),
21     Status VARCHAR(20),
22     FOREIGN KEY (CustomerID) REFERENCES DimCustomer(CustomerID)
23 );
24
25 CREATE TABLE DimBranch (
26     BranchID INT PRIMARY KEY,
27     BranchName VARCHAR(255),
28     BranchLocation VARCHAR(255)
29 );
```


CREATE ETL JOB FOR DIMENSION TABLE

Strategy & Implementation : Building an automated pipeline to standardize and migrate banking master data into structured dimension tables.

Key Actions:

- Data Merging: Integrated multiple sources (Customer, City, State) using Lookup steps.
- Standardization: Applied String Operations to convert names and addresses to UPPERCASE.
- Consistent Naming: Ensured all output fields follow PascalCase conventions.
- Verified Loading: Successfully populated DimCustomer with zero errors, as confirmed by execution logs.



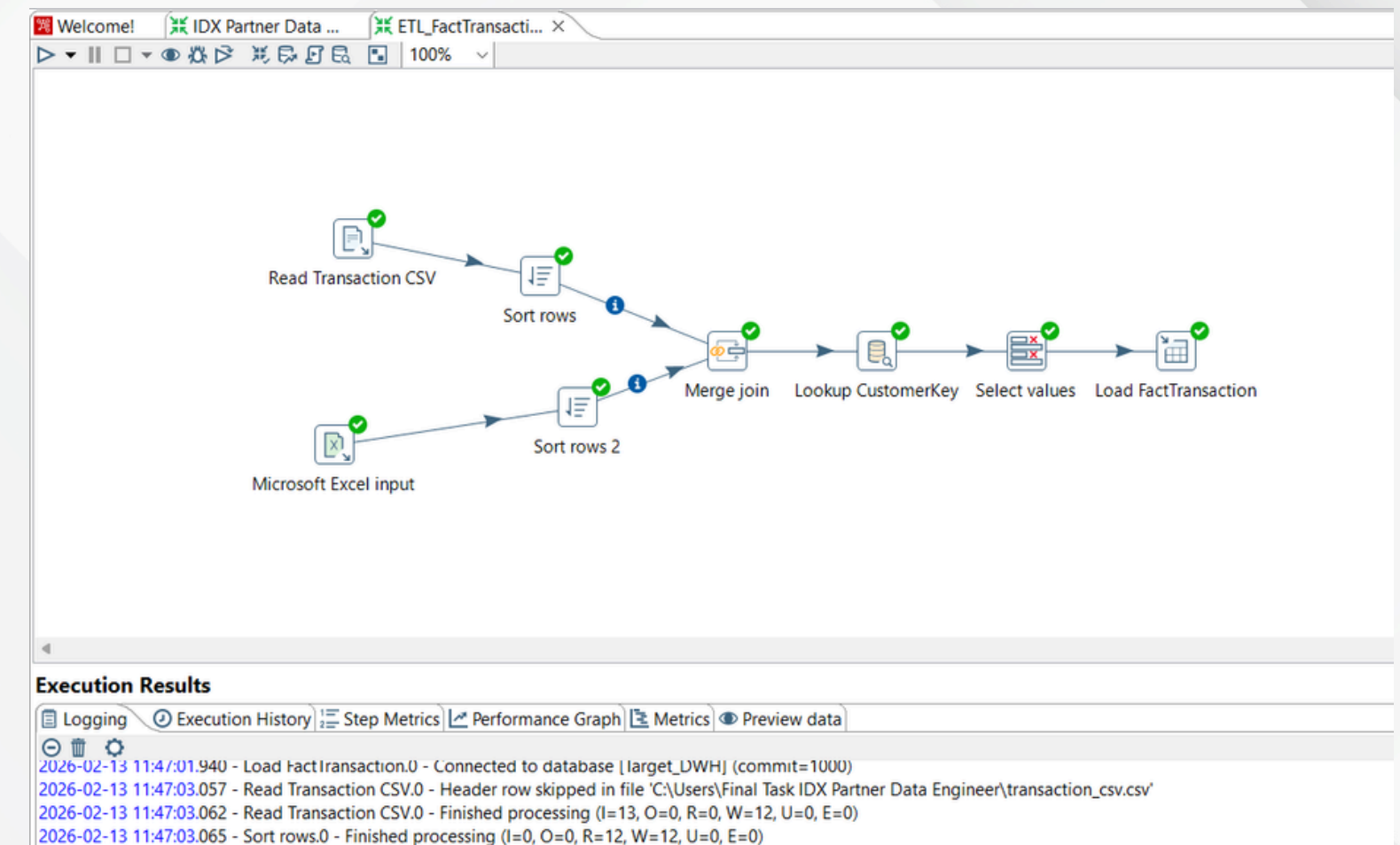
CREATE ETL JOB FOR FACT TABLE

Strategy & Implementation

Integrating high-volume transaction data from disparate sources into a centralized fact table for analytical reporting.

Key Actions:

- **Source Integration:** Unified transaction records from both CSV and Excel file formats.
- **Data Mapping:** Implemented Database Lookup to accurately link transactions with their respective CustomerKey.
- **Flow Optimization:** Utilized Sort Rows and Merge Join steps to ensure seamless data synchronization.
- **Verified Loading:** Successfully populated the FactTransaction table, as confirmed by the “Finished” status in execution logs.



CREATE STORED PROCEDURE

Strategy & Logic Implementation : The goal is to provide the client with automated, parameterized summaries for daily financial reporting and real-time customer balance tracking.

Planned Key Procedures:

- **Daily Transaction Summary:** Developed a procedure with @start_date and @end_date parameters. It aggregates transaction counts and calculates the TotalAmount by summing transactions for each day within the specified range.
- **Balance Per Customer Tracking:** Designed a logic that calculates CurrentBalance by factoring in the initial account balance and transaction history.
- **Conditional Logic:** Implemented CASE WHEN statements where Deposits increase the balance, while other transaction types decrease it.

```
SQLQuery2.s...yosee (79))* ✕ X
1  CREATE PROCEDURE DailyTransaction
2      @start_date DATE,
3      @end_date DATE
4  AS
5  BEGIN
6      SELECT
7          transaction_date AS [Date],
8          COUNT(TransactionID) AS TotalTransactions,
9          SUM(Amount) AS TotalAmount
10     FROM FactTransaction
11     WHERE transaction_date BETWEEN @start_date AND @end_date
12     GROUP BY transaction_date
13     ORDER BY transaction_date ASC;
14 END;
15 GO
```

```
SQLQuery2.s...yosee (79))* ✕ X
1  CREATE PROCEDURE BalancePerCustomer
2      @name VARCHAR(255)
3  AS
4  BEGIN
5      SELECT
6          c.CustomerName,
7          a.AccountType,
8          a.Balance AS InitialBalance,
9          (a.Balance + SUM(CASE WHEN f.transaction_type = 'Deposit' THEN f.Amount
10                                ELSE -f.Amount END)) AS CurrentBalance
11     FROM DimCustomer c
12     JOIN DimAccount a ON c.CustomerID = a.CustomerID
13     JOIN FactTransaction f ON a.AccountID = f.AccountID
14     WHERE c.CustomerName LIKE '%' + @name + '%'
15           AND a.Status = 'active'
16     GROUP BY c.CustomerName, a.AccountType, a.Balance;
17 END;
18 GO
```


PROJECT CONCLUSION & IMPACT

DATA INTEGRATION & GOVERNANCE

Successfully unified fragmented banking data into a centralized Star Schema. By enforcing PascalCase and standardized cleansing, the warehouse ensures a reliable and consistent Single Source of Truth.

AUTOMATION & EFFICIENCY

Automated ETL pipelines in Pentaho replaced manual processing with an instantaneous, production-ready workflow. This optimization eliminates human error and provides scalability for growing transaction volumes.

ANALYTICAL READINESS

Optimized Stored Procedures enable real-time tracking of balances and daily transaction summaries. This analytical layer is fully prepared for BI integration, empowering rapid, data-driven decision-making.

THANK YOU

